

Auditing Polish Hate Speech Detection Models with Functional Tests and Counterfactual Name Cues

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1 Introduction

Automatic hate speech detection is an important NLP task because **online platforms, moderation systems, and researchers increasingly rely on machine learning models to identify harmful content**. However, high aggregate performance on a standard held-out test set does not necessarily mean that a model behaves reliably in realistic or socially sensitive cases. Hate speech classifiers may overfit to surface-level cues such as profanity, group names, or slurs, while failing on more subtle cases such as negation, counter-speech, implicit derogation, or spelling variation. This problem is especially important for languages other than English, where fewer datasets and diagnostic evaluations are available.

In this project, we focus on Polish hate speech and harmful content detection. This particular aspect **remains largely under-researched** in current literature. Our main research question is: *How well do small Polish-specific models detect hate speech across different functional categories, and are their predictions sensitive to irrelevant gender-associated name cues?*

More specifically, we ask:

1. How do Polish models perform on the Polish subset of Multilingual HateCheck, especially across categories such as derogation, threats, slurs, profanity, negation, counter-speech, and neutral mentions of protected groups?
2. Can small models trained or fine-tuned on Polish social media data perform better than the multilingual zero-shot setting reported in the Multilingual HateCheck paper?
3. Does adding an author cue such as “Anna:” or “Jan:” before the same comment change the predicted probability of hatefulness?

The NLP task addressed in the project is **binary text classification**: harmful/hateful versus non-harmful/non-hateful content. We will use two Polish-oriented transformer models. The first is `ptaszynski/bert-base-polish-cyberbullying`, a classifier associated with **Polish cyberbullying and hate speech detection** work originating in the PolEval 2019 shared task and later dataset/model releases (Ptaszynski et al., 2019, 2024). This model can be used directly, although its original label space distinguishes non-harmful, cyberbullying, and hate speech/other harmful content. For our binary task, we will collapse cyberbullying and hate speech into one harmful class.

The second model will be based on **TrelBERT, a Polish Twitter encoder** pretrained with the masked language modeling objective on Polish social media text (Szmyd et al., 2023). Since TrelBERT is a language model rather than a classifier, we will fine-tune it on the Polish cyberbullying dataset introduced and re-annotated by Ptaszynski et al. (2024). The dataset contains Polish tweets split into training and test parts, with labels for non-harmful, cyberbullying, and hate speech/other harmful content. We will convert these labels into a binary harmful/non-harmful setup for fine-tuning.

The main evaluation dataset will be **the Polish subset of Multilingual HateCheck** (Röttger et al., 2022). Multilingual HateCheck is a diagnostic benchmark rather than a typical training dataset. It contains controlled test cases grouped by functional categories, allowing us to inspect not only overall accuracy or macro F1, but also specific weaknesses. The original Multilingual HateCheck paper evaluated an XLM-T-based multilingual model fine-tuned on Spanish, Italian, and Portuguese data, leaving Polish as a zero-shot language. **Our goal is to test whether smaller Polish-specific models can achieve stronger results in some Polish categories.** For example, the XLM-T model achieved great results in identifying ”denouncements of hate that quote it” as non-hate speech, however it incorrectly classified most of the ”non-hate expressed using negated hateful statement” as hate speech.

The experimental phase will have two parts:

- First, we will **evaluate both models on the original Polish HateCheck test cases** and report accuracy, macro F1, false positive rate, false negative rate, and performance by functionality.
- Second, we will create **counterfactual variants** of the same Polish HateCheck examples by adding short author-name prefixes, such as masculine-associated and feminine-associated Polish names.

For example, a sentence will be tested in its original form, with a neutral prefix, with a feminine-name prefix, and with a masculine-name prefix. We will measure probability shifts and label flip rates. Ideally, irrelevant name cues should not change the classification of the same comment. Any systematic difference will be interpreted as possible sensitivity to gender-associated author cues, with the limitation that names are imperfect proxies for gender.

2 Related Work

Hate speech detection has been widely studied as a supervised text classification task, but recent work has shown that **standard dataset-level evaluation can hide important model failures**. Röttger et al. (2021) introduced HateCheck, a functional test suite for English hate speech detection models. Instead of only measuring performance on a random test split, HateCheck evaluates whether models handle specific linguistic phenomena such as negation, slurs, threats, profanity, and counter-speech. Röttger et al. (2022) extended this idea to ten languages, including Polish, in Multilingual HateCheck. Their results showed that even a strong multilingual model can perform inconsistently across languages and can be **overly sensitive to keywords and protected group identifiers**.

For Polish, Ptaszynski et al. (2019) introduced an early shared task and dataset for automatic cyberbullying detection in Polish Twitter. The later work by Ptaszynski et al. (2024) provides an expert-annotated dataset for Polish cyberbullying and hate speech, including both binary harmful/non-harmful classification and a three-class setup distinguishing non-harmful content, cyberbullying, and hate speech/other harmful content. This work is directly relevant to our project because it provides the training data for fine-tuning TrelBERT and motivates the use of Polish-specific models.

TrelBERT was introduced by Szmyd et al. (2023) as a pretrained encoder for Polish Twitter. It is designed for social media language, making it a promising candidate for harmful content detection in Polish tweets and short online comments. In contrast to general multilingual models, TrelBERT is adapted to Polish social media during pretraining. Our project investigates whether this domain-specific pretraining helps in functional hate speech evaluation.

Finally, our counterfactual name-prefix experiment is motivated by broader NLP bias research. Prior work has used counterfactual substitutions to test whether model predictions change when demographic cues are altered while the rest of the input remains constant (Kiritchenko and Mohammad, 2018; Maudslay et al., 2019). However, Devinney et al. (2022) warn that gender bias research often relies on simplified gender

proxies such as names or binary categories. Therefore, we will describe our name experiment carefully: we are not measuring gender identity directly, but model sensitivity to gender-associated Polish first names used as author cues.

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